

Part A

Question 1:

a)

$$1573 = 11^2 \times 13$$

Using Proposition 2.3, HB chapter 2 p17;

for $n \geq 2$, $n = p_1^{k_1} p_2^{k_2} \dots p_r^{k_r}$, then

$$\tau(n) = (k_1 + 1)(k_2 + 1) \dots (k_r + 1) = 1573 = 11^2 \times 13$$

Where $\tau(n)$ is the number of positive divisors of n .

$$k_1 + 1 = 13 \implies k_1 = 12 \quad \checkmark$$

$$k_2 + 1 = 11 \implies k_2 = 10$$

$$k_3 + 1 = 11 \implies k_3 = 10$$

Any other allocation either increases the largest exponent or introduces larger primes, producing a larger integer.

But it also means there are fewer distinct prime divisors, so it isn't clear which choice makes the smallest integer without further analysis.

To find the smallest positive integer n with exactly 1573 positive divisors. To minimise n , we assign the largest exponent to the smallest prime, the second largest exponent to the second smallest prime, and so on.

The easiest way is to consider all 4 possibilities for the powers, and then check directly which is the smallest.

Since $1573 = 13 \cdot 11 \cdot 11$, we choose exponents 12, 10, 10

Hence the smallest positive integer n with exactly 1573 positive divisors is

$$n = 2^{12} \cdot 3^{10} \cdot 5^{10} \quad \checkmark$$

3/4

b)

Let n be the positive integer such that

$$8 \mid 7n + \sigma(n) \quad \text{✗}$$

Where $\sigma(n)$ is the sum of divisors function.

If $n = p^2$ (where p is prime), then $p \equiv 7 \pmod{8}$

Proof.

The divisors of p^2 are 1, p and p^2

Using Proposition 1.2, HB Chapter 9 p48;

$$\sigma(n) = \sum_{d|n} d$$

So

$$\sigma(p^2) = 1 + p + p^2 \quad \checkmark$$

As

$$7n + \sigma(n) \equiv 0 \pmod{8}$$

Substituting $n = p^2$ and $\sigma(p^2)$ gives

$$7p^2 + \sigma(p^2) \equiv 0 \pmod{8}$$

$$7p^2 + 1 + p + p^2 \equiv 0 \pmod{8}$$

Simplifying

$$8p^2 + p + 1 \equiv 0 \pmod{8}$$

As $8p^2 \equiv 0 \pmod{8}$, since $8 \mid 8p^2$, we have

$$p + 1 \equiv 0 \pmod{8} \quad \checkmark$$

$$p \equiv -1 \pmod{8} \Rightarrow p \equiv 7 \pmod{8} \quad \checkmark \quad \text{great!}$$

3/3

6/7

□

Question 2:

$$2x^2 + 3x + 4 \equiv 0 \pmod{7}$$

a)**i.**

Using Euler's criterion, HB Chapter 10 p45, we calculate the discriminant,

$$b^2 - 4ac = 3^2 - 4 \cdot 2 \cdot 4$$

$$= 9 - 32$$

$$= -23$$

$$\equiv 5 \pmod{7}$$


Since the congruence has solutions if and only if the discriminant is a quadratic residue modulo 7, we apply Euler's Criterion.

Using Theorem 1.5, HB Chapter 10 pg 45, we check if 5 is a quadratic residue modulo 7.

$$a^{(p-1)/2} \equiv 1 \pmod{p}$$

$$5^{(7-1)/2} \equiv 5^3 \pmod{7}$$

$$\equiv 125 \pmod{7}$$

$$\equiv 6 \pmod{7}$$

$$\equiv -1 \pmod{7}$$


So 5 is a quadratic non-residue modulo 7, and therefore the congruence

$$2x^2 + 3x + 4 \equiv 0 \pmod{7}$$


has no solutions.

ii.

$$2x^2 + 3x + 4 \equiv 0 \pmod{7}$$

Find the discriminant:

$$\begin{aligned} b^2 - 4ac &= 3^2 - 4 \cdot 2 \cdot 4 \\ &\equiv 9 - 32 \pmod{7} \\ &\equiv 9 + 3 \pmod{7} \\ &\equiv 12 \pmod{7} \\ &\equiv 5 \pmod{7} \end{aligned}$$

By Gauss's lemma, Theorem 3.1 Chapter 10 p46 we determine whether 5 is a quadratic residue modulo 7.:

We need the first $\frac{p-1}{2} = 3$ multiples of 5 $\pmod{7}$:

$$1 \cdot 5 \equiv 5 \pmod{7}$$

$$2 \cdot 5 \equiv 3 \pmod{7}$$

$$3 \cdot 5 \equiv 1 \pmod{7}$$

$$S = \{5, 3, 1\} \pmod{7} \quad \checkmark$$

Looking for the values greater than $\frac{p}{2}$, hence $n = 1$

Using Legendre symbol, Definition 2.1 Chapter 10 p45, and Gauss's lemma:

$$\left(\frac{5}{7}\right) = (-1)^1 = -1$$

4/4

Thus there are no solutions to the quadratic congruence $2x^2 + 3x + 4 \equiv 0 \pmod{7}$. 

b)

i.

$$\left(\frac{-48}{229}\right)$$

Theorem 2.2(c), HB Chapter 10 p46

$$\left(\frac{-48}{229}\right) = \left(\frac{-1}{229}\right) \cdot \left(\frac{16}{229}\right) \cdot \left(\frac{3}{229}\right)$$

Calculating each term separately, using standard results

$$\left(\frac{-1}{229}\right) = 1$$

Theorem 2.2(e), HB Chapter 10 p46

$$\left(\frac{16}{229}\right) = 1$$

Theorem 2.2(b), HB Chapter 10 p46

$$\left(\frac{3}{229}\right) = 1$$

Proposition 4.7, HB Chapter 10 p46

Combining these results

$$\begin{aligned} \left(\frac{-48}{229}\right) &= 1 \cdot 1 \cdot 1 \\ &= 1 \end{aligned}$$

So -48 is a quadratic residue modulo 229

ii.

$$\left(\frac{151}{43}\right)$$

$$\begin{aligned} \left(\frac{151}{43}\right) &= \left(\frac{22}{43}\right) \\ &= \left(\frac{2}{43}\right) \cdot \left(\frac{11}{43}\right) \end{aligned}$$

Theorem 2.2(c), HB Chapter 10 p46

Calculating each term separately

$$\left(\frac{2}{43}\right) = -1$$

Proposition 3.2, HB Chapter 10 p46

$$\left(\frac{11}{43}\right) = -\left(\frac{43}{11}\right)$$

Theorem 4.2, HB Chapter 10 p46

$$= -\left(\frac{10}{11}\right)$$

$$= -\left(\frac{2}{11}\right) \cdot \left(\frac{5}{11}\right)$$


Theorem 2.2(c), HB Chapter 10 p46

Calculate each term separately

$$\left(\frac{2}{11}\right) = -1$$

Proposition 3.2, HB Chapter 10 p46

$$\left(\frac{5}{11}\right) = \left(\frac{11}{5}\right)$$

Theorem 4.2, HB Chapter 10 p46

$$= \left(\frac{1}{5}\right)$$

$$\left(\frac{1}{5}\right) = 1$$


Theorem 2.2(b), HB Chapter 10 p46

Combining these results

$$\left(\frac{151}{43}\right) = \left(\frac{22}{43}\right) = \left(\frac{2}{43}\right) \left(\frac{11}{43}\right)$$


$$\left(\frac{2}{43}\right) = -1$$

and

$$\left(\frac{11}{43}\right) = -\left(\frac{10}{11}\right) = -\left(\frac{2}{11}\right) \cdot \left(\frac{5}{11}\right)$$

$$\left(\frac{2}{11}\right) = -1$$

$$\left(\frac{5}{11}\right) = 1$$

So

$$\left(\frac{11}{43}\right) = -\left(\frac{2}{11}\right) \cdot \left(\frac{5}{11}\right) = -((-1)(1)) = 1$$

therefore

$$\left(\frac{151}{43}\right) = \left(\frac{2}{43}\right) \left(\frac{11}{43}\right) = (-1)(1) = -1$$

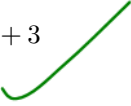


great!
4/4

So 151 is a quadratic non-residue modulo 43

8/8

Therefore the highest common factor is,

$$\text{hcf}(f, g) = x + 3$$


Question 4:

$$f(x) = x^6 + 4x^3 - 2x^2 + 6x + 2$$

a)

Using the Rational Root test, HB Chapter 11 p52, we check for possible rational roots.

$$p \mid 2 \quad q \mid 1$$

The possible rational roots are $\frac{p}{q} = \pm 1, \pm 2$.

$$f(1) = 1 + 4 - 2 + 6 + 2 = 11 \neq 0$$

$$f(-1) = 1 - 4 - 2 - 6 + 2 = -9 \neq 0$$

$$f(2) = 64 + 32 - 8 + 12 + 2 = 102 \neq 0$$

$$f(-2) = 64 - 32 - 8 - 12 + 2 = 10 \neq 0$$

4/4

Therefore $f(x)$ has no rational roots. Hence $f(x)$ has no linear factor over $\mathbb{Q}$.

b)

Using Eisenstein's criterion, HB Chapter 11 p52, we check for irreducibility over $\mathbb{Q}$.

Let $p = 2$.

- $2 \mid 4, -2, 6, 2$
- $2 \nmid 1$
- $2^2 = 4 \nmid 2$

So by Eisenstein's criterion, $f(x)$ is irreducible over $\mathbb{Q}$, hence it has no linear factor over $\mathbb{Q}$, so it has no rational roots.

2/2

6/6

Question 5:**a)****i.**

Suppose;

$$\sqrt{50} = \frac{m}{n}, \quad \text{where } m, n \in \mathbb{Z}^+$$

Show

$$\sqrt{50} = \frac{50n - 7m}{m - 7n}$$

Proof.

Since

$$\sqrt{50} = \frac{m}{n}$$

We have

$$m = n\sqrt{50}$$

$$\frac{m}{n} = \frac{50n - 7m}{m - 7n}$$

This is what you're trying to show
- you don't know it yet. This is a
circular argument.

Cross-multiplying

$$m(m - 7n) = n(50n - 7m)$$

expanding

$$m^2 - 7mn = 50n^2 - 7mn$$

Adding $7mn$ to both sides

$$m^2 = 50n^2$$

(i) If $m = \sqrt{50}n$ then

Taking square roots of both sides

$$\sqrt{m^2} = \sqrt{50n^2}$$

$$m = n\sqrt{50}$$

Dividing both sides by n

$$\frac{m}{n} = \sqrt{50}$$

$$\therefore \sqrt{50} = \frac{m}{n}$$

not what is asked

$$\frac{50n - 7m}{m - 7n} = \frac{50n - 7\sqrt{50}n}{\sqrt{50}n - 7n}$$

(substituting)

$$= \frac{50 - 7\sqrt{50}}{\sqrt{50} - 7}$$

(cancelling n)

$$= \frac{(50 - 7\sqrt{50})(\sqrt{50} + 7)}{(\sqrt{50} - 7)(\sqrt{50} + 7)}$$

(clearing denominator)

$$= \frac{(50 - 7\sqrt{50})(\sqrt{50} + 7)}{1}$$

$$= \sqrt{50}$$

(calculation)

□

0/2

ii.

Suppose;

$$\sqrt{50} = \frac{m}{n} = \frac{50n - 7m}{m - 7n}$$

By Fermat's infinite descent, HB Chapter 12 p53, to show that $\sqrt{50}$ is irrational.

Proof.

Assuming $\sqrt{50}$ is rational, then $\frac{m}{n}$ is in lowest terms

With $m, n \in \mathbb{Z}^+$ and $\text{hcf}(m, n) = 1$

From part (a), we have

$$\sqrt{50} = \frac{50n - 7m}{m - 7n}$$

We define $m' = 50n - 7m$, $n' = m - 7n$

Since $m, n \in \mathbb{Z}$, it follows that $m', n' \in \mathbb{Z}$

Since

$$7 < \sqrt{50} < 8$$

$$m' = 50n - 7m = n(50 - 7\sqrt{50}) > 0$$

and

$$n' = m - 7n = n(\sqrt{50} - 7) > 0$$

So $m', n' \in \mathbb{Z}^+$

Also, using $7 < \sqrt{50}$

$$m' = n(50 - 7\sqrt{50}) < n(50 - 49) = n < m$$

and since $m = n\sqrt{50} > n$, $n' < n$ *I don't follow.*

$$\frac{m'}{n'} = \frac{m}{n}$$

but with $m' < m$ and $n' < n$, this contradicts our

assumption that $\frac{m}{n}$ is in lowest terms.

lowest terms means that m and n have no common factor, and you have not contradicted that.

2/3

Repeating this process would produce an infinite decreasing sequence of positive integers which is impossible. So Our initial assumption leads to a contradiction.

this is the right thing to say in an infinite descent proof.

Therefore $\sqrt{50}$ is irrational.

□

b)**i.**

$$578 = 2 \times 289$$

since;

$$289 = 17^2$$

So we get our first representation as the sum of two squares;

$$578 = 17^2 + 17^2$$

$$23^2 = 529$$

Subtracting from 578 gives;

$$578 - 23^2 = 49$$

$$49 = 7^2$$

So our second representation as the sum of two squares is;

$$578 = 23^2 + 7^2$$

Thus the two representations of 578 as the sum of two squares are: First representation:

$$578 = 17^2 + 17^2$$

Second representation:

$$578 = 23^2 + 7^2$$

3/3

ii.

$$1715 = 2 \times 7^3$$

$$7 \equiv 3 \pmod{4}$$

2/2

Using Proposition 4.13 Chapter 12 p57, since $7 \equiv 3 \pmod{4}$ appears with odd exponent, 1715 cannot be expressed as the sum of two squares.

7/10

Question 6:

$$R = \{a_0 + a_1x + a_2x^2 + a_3x^3 + \dots + a_nx^n : a_0, a_1, \dots, a_n \in \mathbb{Z}_{59}, a_1 = 0, n \geq 0\}$$

a)

Proof. Using the subring criterion, Lemma 1.8 HB Chapter 11 p48,

Closure under subtraction;

Let

$$\begin{aligned} f(x) &= a_0 + a_2x^2 + a_3x^3 \dots + a_nx^n \\ g(x) &= b_0 + b_2x^2 + b_3x^3 \dots + b_nx^n \end{aligned}$$

Include the final term, otherwise the notation implies that the sum goes on forever. If you are worried about the degrees being different, you can take n to be the larger of the degrees of s and t , and set the coefficients of any unwanted powers to be 0.

Then

$$f(x) - g(x) = (a_0 - b_0) + (a_2 - b_2)x^2 + (a_3 - b_3)x^3 \dots$$

Since $a_i, b_i \in \mathbb{Z}_{59}$, then $a_i - b_i \in \mathbb{Z}_{59}$

Also, the coefficient of x is $0 - 0 = 0$

So $f(x) - g(x) \in R$

Closure under multiplication;

In order to show closure under multiplication of a polynomial with no x term we consider the product of multiplying two polynomials in R to create a x term.

The coefficient of x in a product comes from terms where the degrees add to 1. This can happen with terms of degree $1 + 0$ or $0 + 1$.

As the coefficient of x in both polynomials is 0, the coefficient of x in the product will also be 0.

Thus the coefficient of x in $f \cdot g$ is 0 so $f \cdot g \in R$.

Since $R \subseteq \mathbb{Z}_{59}[x]$, we apply the subring criterion.

Hence, R is a subring of $\mathbb{Z}_{59}[x]$.

□

Since $1 = 1 + 0x + 0x^2 + \dots \in R$, R contains the multiplicative identity.

Units in a polynomial ring over a field are the non-zero constants,
Definition 1.14 Chapter 11 p48.

If $u \in R$ is a unit in R , then it is also a unit in $\mathbb{Z}_{59}[x]$ hence must be a non-zero constant, so $U(R) = \mathbb{Z}_{59}^*$

Therefore the units in R are given by;

$$U(R) = \{a_0 : a_0 \in \mathbb{Z}_{59}^*\}$$

where

$$\mathbb{Z}_{59}^* = \{1, 2, 3, \dots, 58\}$$

4/4

b)

By Lemma 2.6 chapter 11 p49,

In order for x^2 to be reducible in R there must exist polynomials $p(x), q(x) \in R$ such that

$$x^2 = p(x) \cdot q(x)$$

This leaves three possibilities for the degrees of $p(x)$ and $q(x)$:

- $\deg(p) = 0, \deg(q) = 2$
- $\deg(p) = 2, \deg(q) = 0$
- $\deg(p) = 1, \deg(q) = 1$

You need to justify this step by saying "since 59 is prime, $\mathbb{Z}_{59}$ is a field, and hence Lemma 2.6 HB pg 49 applies."

Note that the degree formula does not always hold: e.g. in $\mathbb{Z}_6[x]$ we have $(2x+1)(3x+1) = 6x^2+5x+1 = 5x+1$, so letting $s = 2x+1$, $t=3x+1$, we get $\deg(st) = 1$, but $\deg(s) + \deg(t) = 2$.

The first two possibilities are not valid as the constant polynomials are units in R and hence cannot give a non-trivial factorisation.

The third possibility is not valid since no polynomial in R has a non-zero coefficient of x .

2/3

c)

$$x^2 \nmid x^3 \text{ in } R$$

Proof. Assume for contradiction that $x^2 \mid x^3$ in R . Then there exists $h(x) \in R$ such that

$$x^3 = x^2 \cdot h(x).$$

prove this - see below

~~In $\mathbb{Z}_{59}[x]$, this forces $h(x) = x$. But $x \notin R$ because elements of R have zero coefficient of x . This contradiction shows $x^2 \nmid x^3$ in R . $\square$~~

Prove x^2 is not a prime element in R .

2/3

Proof.

For x^2 to be prime in R , whenever $x^2 \mid f(x) \cdot g(x)$, then $x^2 \mid f(x)$ or $x^2 \mid g(x)$.

Consider the polynomials

$$x^2 \mid x^6 \text{ since } x^6 = x^2 \cdot x^4 \text{ both } x^4, x^6 \in R$$

But

$$x^6 = x^3 \cdot x^3 \text{ and } x^2 \nmid x^3$$

good

An element p is prime if whenever $p \mid ab$ then $p \mid a$ or $p \mid b$

Hence x^2 is not a prime element in R . $\square$

8/10

$\mathbb{Z}_{59}$ is a field so we can use Lemma 2.6. of Chapter 11, HB p49.

Suppose that x^2 divides x^3 , so there exists $f \in R$ such that $f \cdot x^2 = x^3$. By Chapter 11, p128, Lemma 2.6, HB p49 we must have $\deg(f) = 1$. But this implies that the coefficient of x in f is non-zero, which is impossible since $f \in R$.

Question 7:

PA Paul Allen
25 Dec 2025 23:34

I have been going through this question and think I can solve it using algebra. But I have been looking ahead in the work and it seems there is possibly other ways to solve part a) but it isn't covered until book E. I haven't gone through that work properly yet so I'm not sure if I am even correct, I was wondering if we are expected to be using stuff ahead of where we are at?

Has anybody else found alternative solutions to previously done questions as we get further into the material?

Any advise would be welcome before I get ahead of myself.

Thanks

Paul

2/2

Part B

Question 8:

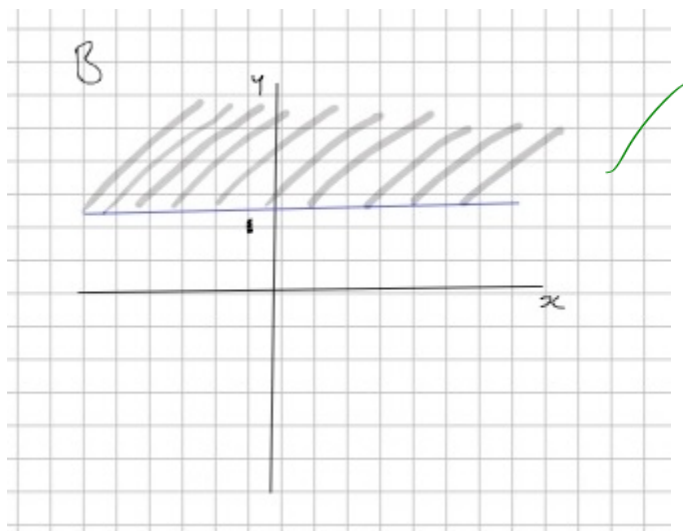
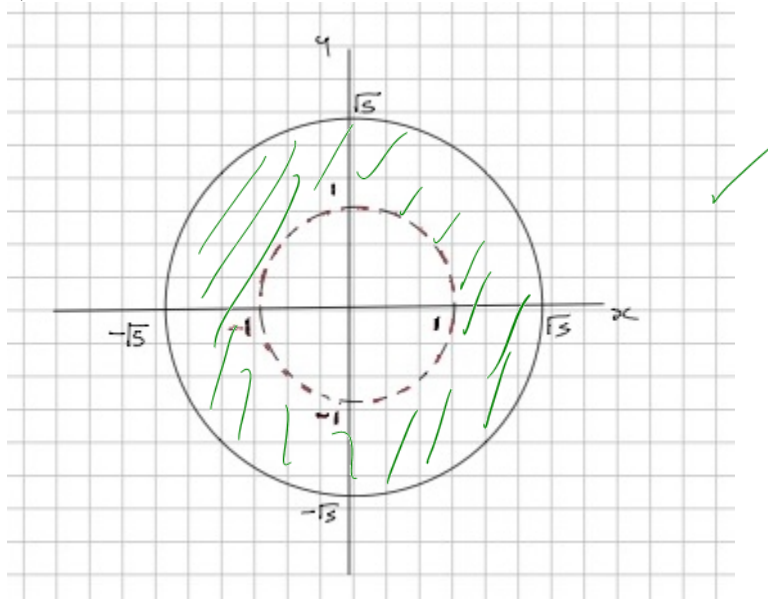
Question 8 from the TMA03 assignment.

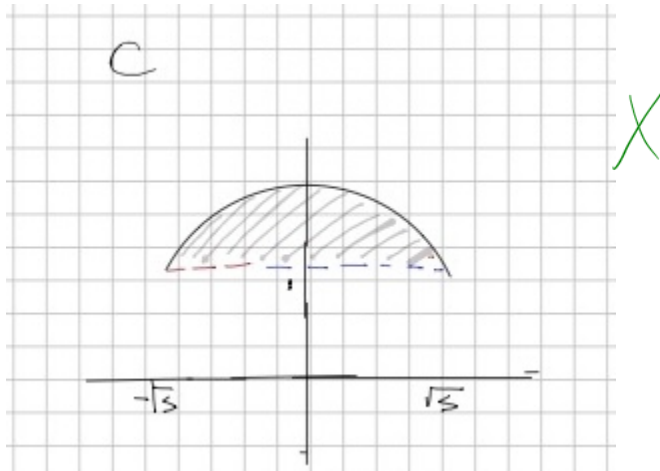
$$A = \{(x, y) \in \mathbb{R}^2 : 1 < x^2 + y^2 \leq 5\}$$

$$B = \{(x, y) \in \mathbb{R}^2 : y \geq 1\}$$

$$C = A \cap B$$

a)





5/6

b)

$$\left(0, 1 + \frac{1}{n}\right)_{n=1}^{\infty}$$

By Definition 3.4 HB Chapter 13 p67,

The first component is a constant sequence $\lim_{n \rightarrow \infty} 0 = 0$ and constants sequences converge to the constant value.

The second component is $1 + \frac{1}{n}$.

Splitting this into two limits;

$$\lim_{n \rightarrow \infty} 1 = 1$$

Again it is a constant sequence, so it converges to 1.

$$\lim_{n \rightarrow \infty} \frac{1}{n} = 0$$

As n approaches infinity, $\frac{1}{n}$ approaches 0.

The question tells you to use Def 3.4, but in fact you are using Theorem 3.7 to split the sequence up into its coordinates, rather than applying the definition itself.

If the question specifies a method you must use it to get the marks.

Combining these results;

$$\lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right) = 1 + 0 = 1$$

Therefore, by Definition 3.4 HB Chapter 13 p67, the sequence

$$\left(0, 1 + \frac{1}{n}\right)_{n=1}^{\infty}$$

converges to

$$(0, 1)$$

To test the limit point $(0, 1) \in A$,

$$1 < x^2 + y^2 \leq 5$$

$$1 < 0^2 + 1^2 \leq 5$$

$$1 < 1 \leq 5$$

This is false, so $(0, 1) \notin A$

To test the limit point $(0, 1) \in B$,

$$y \geq 1$$

$$1 \geq 1$$

This is true, so $(0, 1) \in B$

3/5

8/11

Hence limit point $(0, 1) \notin A$ but $(0, 1) \in B$,

Therefore $(0, 1) \notin C$.

Question 9:

Question 10 from the TMA03 assignment.

Let

$$R = (\mathbb{Z}, \oplus, \odot)$$

Where

$$\oplus = a + b + 1$$

and

$$\odot = ab + a + b$$

a)

Proof.

R1 and R6 Closure of $\oplus$ and $\odot$;

$$a, b, 1 \in \mathbb{Z}$$

Therefore, because of closure under addition in $\mathbb{Z}$,

$$a + b + 1 \in \mathbb{Z}$$

So $a \oplus b \in \mathbb{Z}$

$$ab, a, b \in \mathbb{Z}$$

Therefore, because of closure under multiplication and addition in $\mathbb{Z}$,

$$ab + a + b \in \mathbb{Z}$$

So $a \odot b \in \mathbb{Z}$

R2 and R7 Associativity of $\oplus$ and $\odot$;

For $\oplus$,

$$(a \oplus b) \oplus c = (a + b + 1) + c + 1$$

$$= a + b + c + 2$$

And

$$\begin{aligned} a \oplus (b \oplus c) &= a + (b + c + 1) + 1 \\ &= a + b + c + 2 \end{aligned}$$

$$\text{So } (a \oplus b) \oplus c = a \oplus (b \oplus c)$$

For $\odot$,

$$\begin{aligned} (a \odot b) \odot c &= (ab + a + b) \odot c \\ &= (ab + a + b)c + (ab + a + b) + c \\ &= abc + ac + bc + ab + a + b + c \end{aligned}$$

And

$$\begin{aligned} a \odot (b \odot c) &= a \odot (bc + b + c) \\ &= a(bc + b + c) + a + (bc + b + c) \\ &= abc + ab + ac + a + bc + b + c \end{aligned}$$

$$\text{So } (a \odot b) \odot c = a \odot (b \odot c)$$

R3 Additive identity;

Let e be the additive identity, then

$$\begin{aligned} a \oplus e &= a \\ a + e + 1 &= a \\ e + 1 &= 0 \\ e &= -1 \end{aligned}$$

Your logic flows in the wrong direction. You need to start by defining $e = -1$, and then check that it satisfies the definition of an additive identity.

So the additive identity is -1

R8 Multiplicative identity;

Let f be the multiplicative identity, then

$$a \odot f = a$$

$$af + a + f = a$$

$$af + f = 0$$

$$f(a + 1) = 0$$

Same comment: logic flows in the wrong direction here. Simply define $f=0$, and then check that it works.

For a given a this can happen if $f = 0$ or $a = -1$

But f must work for all a , and we can choose $a \neq -1$ e.g. $a = 0$

$$a + 1 \neq 0$$

$$0 + 1 \neq 0$$

$$1 \neq 0$$

Hence $f = 0$

R4 Additive inverse;

Let g be the additive inverse of a , then

$$a \oplus g = -1$$

$$a + g + 1 = -1$$

$$g + a = -2$$

$$g = -a - 2$$

Same comment: logic flows in the wrong direction.

So the additive inverse of a is $-a - 2$ ✓

R5 Commutativity of $\oplus$;

$$a \oplus b = a + b + 1$$

$$= b + a + 1$$

$$= b \oplus a$$

So $a \oplus b = b \oplus a$

R9 Distributive law;

$$a \odot (b \oplus c) = a \odot (b + c + 1)$$

$$= a(b + c + 1) + a + (b + c + 1)$$

$$= ab + ac + a + a + b + c + 1$$

$$= ab + ac + 2a + b + c + 1$$

And

$$(a \odot b) \oplus (a \odot c) = (ab + a + b) \oplus (ac + a + c)$$

$$= (ab + a + b) + (ac + a + c) + 1$$

$$= ab + ac + a + a + b + c + 1$$

$$= ab + ac + 2a + b + c + 1$$

So $a \odot (b \oplus c) = (a \odot b) \oplus (a \odot c)$

R10 Commutativity of $\odot$;

$$a \odot b = ab + a + b$$

$$= ba + b + a$$

$$= b \odot a$$

So $a \odot b = b \odot a$

5/5

Therefore, R is a commutative ring with identity.

□

b)

By Definition 2.8 HB Chapter 12 p54, we prove R is an integral domain;

Proof.

From part a), we know R is a commutative ring with identity.

No zero divisors;

Let $a, b \in R$ such that $a \odot b = 0$, then

You need the additive identity here, which is -1 not 0

$$\underline{ab + a + b = 0}$$

adding 1 to both sides,

$$ab + a + b + 1 = 1$$

Factorising the LHS,

$$(a + 1)(b + 1) = 1$$

Since $a, b \in \mathbb{Z}$, the only way for the product to be 1 is if both $(a + 1) = 1$ and $(b + 1) = 1$

$$a + 1 = 1$$

$$b + 1 = 1$$

Therefore,

$$a = 0$$

$$b = 0$$

1/3

So the only way for $a \odot b = 0$ is if $a = 0$ or $b = 0$

Therefore, R has no zero divisors, and is an integral domain. $\square$

c)

By Definition 1.14 HB Chapter 11 p48, we have already shown **R1-R10**, so we also need to find a multiplicative inverse for every non-zero element in R .

Multiplicative inverse;

Let $a \in R$ and $a \neq 0$, and let h be the multiplicative inverse of a , then

$$a \odot h = 0$$

$$ah + a + h = 0$$

$$ah + h = -a$$

$$h(a + 1) = -a$$

$$h = \frac{-a}{a+1}$$

So the multiplicative inverse of a is $\frac{-a}{a+1}$

However, for h to be in R , $\frac{-a}{a+1}$ must be an integer, which is not the case for all $a \in R$

For example, if $a = 1$, then $h = \frac{-1}{2} \notin \mathbb{Z}$

good
3/3

Therefore, not every non-zero element in R has a multiplicative inverse, and so R is not a field.

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